Student Handout

Name: _____ **Date:** _____ **Class:** _____

MyDesign Portfolio

Element D Initial Template

Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision making on which design to include, ideally with a design matrix. Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document. Generate a design blueprint with enough details that someone else not familiar with your project could also build your design.

You will then proceed forward to build and test this design, returning to Element D afterwards when you are ready to create more design ideas. Fill out each of these steps again.

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

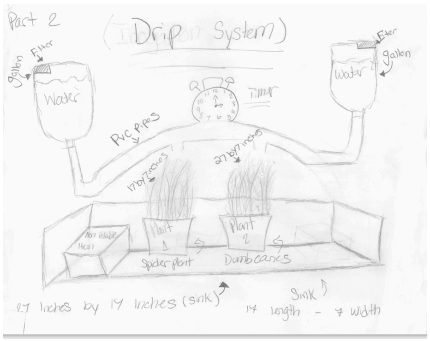
This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

Design Idea #1

Name: _____ **Date:** _____ **Class:** _____

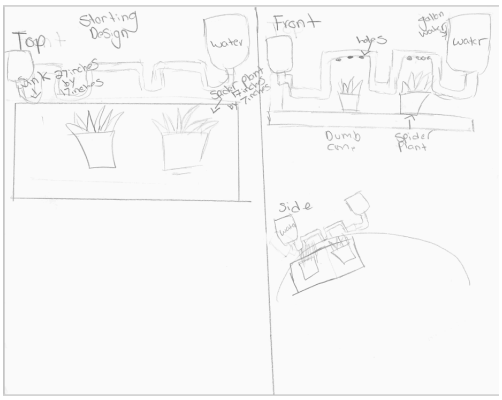
Date:3/18/2025

Description: This first concept design is an indoor drip irrigation system made from PVC pipes shaped into an arch. Two large gallon containers of water are placed on each side, acting as reservoirs. The water flows through the PVC pipes, which have small holes evenly spaced along the arch to drip water onto plants below. A timer controls when the water is released, making sure the plants are watered at the right times while saving water. Materials include PVC pipes, gallon containers, a timer, and drip nozzles if needed.



Sink: width 26 inches 17 inches length
Plant 1: 27 inches length 7 width
Plant 2: 17 length 7 width

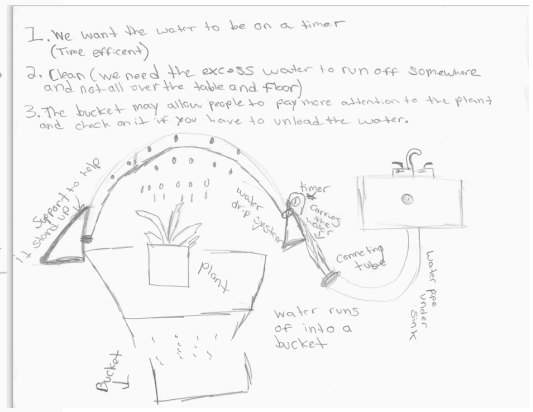
Dumb canes - 0.8 cups of water every 9 days
Spider plant - Once a week
Amount: Around 1/4 to 1/2 cup of water



SHOPPING LIST:

- water storage tank (w/ stand) 2 #(bottles prototype)
- PVC pipe multipule
- timer
- water FILTER
- Flexible plastic Tubing
- Ziptie

pepion, russell, ameira alfonso



Name: _____ **Date:** _____ **Class:** _____

Design #2

Date: 3/20/2025

Description:

A modified version of Idea #1, using IV needles attached to plastic tubing to precisely deliver water to each plant. Water moves from the reservoir into tubing connected to the IV needles, ensuring targeted watering. No large PVC arch is used; instead, the system is flat and customizable to any arrangement of plants.

Materials: IV needles, clear plastic tubing, IV bags , Pvc pipes.

Design Idea #3

Date: 3/22/2025

Description:

A vertical drip system made from stacked PVC frames where plants are suspended vertically with water dripping from the top down through each level. Each level receives decreasing amounts of water. A solar-powered timer controls watering frequency to save energy.

Materials: Thin PVC pipes, gallon containers, solar timer, drip emitters, and plastic frames.

Drawing:

After comparing the ideas, Design Idea #2 (the IV needle and tubing system) scored highest. It is the simplest to set up, the most water-efficient, and easiest to maintain. Although slightly more expensive due to the cost of needles and tubing, its advantages outweigh the others.

Design Chosen: Design Idea #2: IV Needle Drip System

Name: _____ **Date:** _____ **Class:** _____

Plan of action

1 Research drip systems and gather materials (tubing, IV needles, containers)

2 Cut and assemble tubing system with needles and attach to containers

3 Test water flow rates and adjust needle placements

4 Set up final system above plants and adjust timer settings

5 Monitor plant hydration and make system tweaks as needed

Detailed Task Breakdown:

- Research optimal drip rates for indoor plants
- Measure and cut tubing sections based on plant layout
- Attach IV needles securely to tubing ends
- Insert tubing into Iv bags
- Create support system for tubing if needed

- Test for leaks and flow consistency
- Adjust as needed

Name: _____ **Date:** _____ **Class:** _____

Design Blueprint

- Cut plastic tubing to the needed lengths using scissors
- Attach IV needles tightly to the tubing ends
- Seal tubing into gallon containers (use hot glue)
- Suspend tubing above plants using small hooks or rods

Item	Cost Each	Quantity Needed	Total Cost
IV Bags (1-liter each)	$4	2	$8
Plastic tubing (10 ft)	$5	1	$5
IV needles	$0.50	10	$5
Mechanical timer	$10	1	$10
Hooks and support rods	$1	5	$5

Name: _____ **Date:** _____ **Class:** _____

Total Cost **$33**

Criterion	Weight	Idea #1	Idea #2	Idea #3
Simplicity of Setup	30%	4	5	3
Water Efficiency	25%	3	5	4
Cost	20%	4	3	2
Maintenance Difficulty	15%	3	5	2
Material Availability	10%	5	5	3
Total Score		3.65	**4.75**	2.95

Name: _____ **Date:** _____ **Class:** _____

	weight	Option 1	Option 2	Option 3	Option 4
Criterion B					
Criterion C					
Criterion D					
Criterion E					
score					

Design Blueprint *(D4)*

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Date:
Sketch:
Manufacturing Methods:
    Step 1:
    Step 2:
    Step 3:
Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**

Name: _____ **Date:** _____ **Class:** _____

Total Cost			

Plan of Action *(D3)*

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.